

TASK GROUP INSTRUCTION

AIM REV 20.11.4.0 (OCTOBER 2020)

TGI NO.:	COMPONENT NAME:	TYPE:	PHASE:	REPEAT:	CHANGE:
	FWD INTAKE	AY	A	0	0
SHIP/HULL NO.:	SHIP/HULL NAME:	SYSTEM:		PROJ/WO NO:	CASREP CAT:
DDG-76	USS HIGGINS			KB3	
CU PHASE TITLE:					
F/O – INSPECT, REPAIR, AND PRESERVE IN NO. ONE GTM AND SSGTG NO. ONE INTAKE					
PREPARING ACTIVITY:					
PREPARED BY:	CODE	PHONE	DATE		
MISAKI KANEHIRA	231A	315-243-9883	19-DEC-2025		
APPROVED BY:	CODE	PHONE	DATE		
YUUJI NAKAMURA	231A	315-243-6041	19-DEC-2025		
CONCURRED BY:	CODE	PHONE	DATE		
CONCURRED BY:	CODE	PHONE	DATE		
CONCURRED BY:	CODE	PHONE	DATE		
CONCURRED BY:	CODE	PHONE	DATE		
REASON FOR WORK/NEED FOR CHANGE:					
CHANGE INSTRUCTIONS:					
SPECIAL REQUIREMENTS:					
SURFMEPP This work affects a fire zone boundary. Fabrication and/or staging of a non-flammable closure is required					

WORK CERTIFICATION SIGNATURES					
COMPLETION OF WORK REVIEW:		BADGE NO.:	SHOP/CODE:	PHONE:	DATE:
ACCEPTANCE OF COMPLETED WORK					
PHYSICAL INSPECTION	PHYSICAL INSPECTION COMPLETED:	BADGE NO.:	SHOP/CODE:	PHONE:	DATE:
REQUIRED: ☐ YES ☒ NO					
RECORDS REVIEW:		BADGE NO.:	SHOP/CODE:	PHONE:	DATE:



LIST OF EFFECTIVE PAGES									
SHT NO.	CHG NO.	SHT NO.	CHG NO.	SHT NO.	CHG NO.	SHT NO.	CHG NO.	SHT NO.	CHG NO.
1-6									
J1-2									
M-1									

1. SCOPE/PURPOSE

1.1 Title: Engine Room No. One Gas Turbine Module (GTM) and Ship Service Gas Turbine Generator (SSGTG) No One Dirty Side Intake; inspect, repair, and preserve

1.2 Location of Work:

1.2.1 Intake Engine Room No. One, Port (Dirty Side) (02-174-0-Q)

1.2.2 Intake Engine Room No. One, Starboard (Stbd) (Dirty Side) (02-174-1-Q)

1.2.3 SSGTG Intake (Dirty Side) (03-174-0-Q)

1.2.4 SSGTG Intake/ Exhaust Trunk (Dirty Side) (02-161-2-Q)

1.2.5 SSGTG Intake Trunk (2-161-2-Q)

1.2.6 Tomahawk Equipment Room (2-153-2-C)

1.2.7 SSGTG Intake/ Exhaust Trunk (Clean Side) (02-161-2-Q)

1.2.8 Intake Engine Room No. One, Port (Clean Side) (02-174-0-Q)

1.2.9 Intake Engine Room No. One, Stbd (Clean Side) (02-174-1-Q)

1.2.10 Auxiliary Machinery Room No. One (4-126-0-E)

1.2.11 Engine Room No. One (4-174-0-E)

1.3 Identification:

1.3.1 Quantity (One EA), Dirty Side of No. 1A GTM Intake in 02-174-1-Q

1.3.2 Quantity (One EA), Dirty Side of No. 1B GTM Intake in 02-174-0-Q

1.3.3 Quantity (One EA), Dirty Side of No. One SSGTG Intake in 03-174-0-Q

1.3.4 Quantity (One EA), Dirty Side of Intake/Exhaust Trunk in 02-161-2-Q

1.4 Security Classification:

1.4.1 Space: 1.2.6

1.5 Initial Condition: None.

1.6 Contractor work start date: 29/JUN/2026

1.7 Contractor work completion date: 8/DEC/2026

2 TGI NO.:

SHIP:

FWD INTAKE	AY	A	0
DDG-76	CHG/VER:		0/0

2. **REFERENCES:**

REF. NO.	DOCUMENT /DRAWING NUMBER	DOCUMENT/DRAWING TITLE	REV	CHG NO.	COPY REQ'D (Y/N)	SOURCE DOC (Y/N)	PARA	SEC	PG	CHP
1	NSI FY-26	NAVSEA Standard Items	-	1	N					
2	LI FY-26	Local Standard Items	-	ORG	N					
3	MP LM23-KB312388-EM01ZCJ5	1A /1B GTM & NR.1 SSGTG INTK SYS FOD PRTN FOR PARTIALLY RPR GAS TURB FWD INTK SP	-		Y				AL L	
4	DWG 150-6218956	UNIT STRL ARR DWG ASSY UNIT 4220	AB		Y				3-9,5 C-9	
5	DWG 150-6218957	UNIT STRUCTURAL ARR DWG ASSY UNIT 4230	W		Y				SE E N O T E	
6	DWG 150-6218962	UNIT STRL ARR DWG ASSY UNIT 4310	V		Y				3-5	
7	DWG 150-6218963	DDG-51 CLASS UNIT STRL ARR DWG ASSY UNIT 4320	AA		Y				SE E N O T E	
8	TM SYSTEMS & SPECIFICAT IONS	SSPC Painting Manual Volume 2	-		N					
9	TM T9074-AS-GIB-010/271	Requirements for Nondestructive Testing Methods	1		N					
10	TM S9086-DA-STM-010/CH-100	Hull Structures	4		N					
11	MIL MIL-STD-2035	Nondestructive Testing Acceptable Criteria	A		N					

3. **APPENDICES:**

- 3.1. Appendix "J" TGI Job Summary Information Sheet
- 3.2. Appendix "M" TGI Material List
- 3.3. Appendix "D" Deficiency Form

4. **GENERAL REQUIREMENTS:**

4.1 Not Applicable.

5. **PROCEDURE:**

5.1 Accomplish the requirements of 2.3.

(V)(G) "GAS TURBINE MAINTENANCE PROCEDURE"

5.2 Accomplish a visual inspection of the current ship condition, listed in 1.3.1 through 1.3.4 and located in 1.2.1 through 1.2.4, using 2.4 through 2.7 for guidance.

5.2.1 Ensure to inspect each deck, bulkhead, overhead surface, steel channel, stiffener, penetration, pipe hanger and vibration damping.

5.2.2 Mark each corroded, damaged and/or deteriorated area.

3 TGI NO.:

SHIP:

FWD INTAKE	AY	A	0
DDG-76	CHG/VER:		0/0

5.2.3 Submit one legible copy, in hard copy or approved transferable media, of a report with a sketch listing results of the requirements of 5.2 to the SUPERVISOR.

5.3 Power tool clean to bare metal each location reported in 5.2.3, located in 1.2.1 through 1.2.4, to include each exposed longitudinal, vertical, and traverse stiffener, for up to 1,468 square feet. Accomplish each requirement of Surface Preparation Specification SSPC-SP 11 of 2.8, as designated by the SUPERVISOR.

5.4 Accomplish a visual inspection of each surface power tool cleaned to bare metal in 5.3, listed in 1.3.1 through 1.3.4, located in 1.2.1 through 1.2.4 for structural integrity, deterioration, pitting, each crack, and area of damage or distortion, using 2.4 through 2.7 for guidance.

5.4.1 Mark each corroded, damaged and/or deteriorated area.

5.4.2 Submit one legible copy, in hard copy or approved transferable media, of a report with a sketch listing results of the requirements of 5.4 to the SUPERVISOR.

5.5 Accomplish each ultrasonic thickness measurement (UTM) up to 100 locations in each location reported in 5.4.2, listed in 1.3.1 through 1.3.4, located in 1.2.1 through 1.2.4 in accordance with Paragraph 6.7 of 2.9 and 2.10, as designated by the SUPERVISOR, using 2.4 through 2.7 for guidance. The acceptance criteria must be in accordance with 2.11.

5.5.1 Submit one legible copy, in hard copy or approved transferable media, of a report listing results of the requirements of 5.5 to the SUPERVISOR.

5.5.2 Ensure a sketch showing each location of each UTM reading, original material thickness, existing material thickness, and percentage of material thickness reduction.

5.6 Accomplish the following repairs reported in 5.2.3, 5.4.2 and 5.5.1, listed in 1.3.1 through 1.3.4, located in 1.2.1 through 1.2.4 as designated by the SUPERVISOR, using 2.4 through 2.7 for guidance. See 5.9.5 and 5.9.6.

5.6.1 Base metal buildup, a total of 50 square inches.

5.6.1.1 Ensure each base metal buildup area does not exceed 2 inches in diameter.

5.6.1.2 Estimate the base metal buildup to be one quarter inch thick for pricing purposes.

5.6.1.3 Fill each local pit and groove with metal prior to the application of each principal build-up layer.

5.6.1.4 Each surface must be ground smooth to metal.

5.6.2 Remove existing and install new, a total of 15 square feet of deck, overhead, and bulkhead.

5.6.3 Remove existing and install new, a total of 10 linear feet of longitudinal, vertical, and transverse stiffener.

5.6.4 Vee out and weld a total of 4 linear feet of cracked, split, broken or damaged intake weld.

5.6.5 Chip and grind each surface flush in way of each repair.

5.6.6 Template exact size, configuration, and location from each existing shipboard condition.

5.6.7 Accomplish the requirements of 009-012 of 2.1, including Table 2, Columns A and D, Lines One through 7.

5.6.8 Accomplish the requirements 009-025 of 2.1 for an air hose test of each new weld seam. Allowable leakage: None.

FWD INTAKE	AY	A	0
DDG-76	CHG/VER:	0/0	

5.7 Accomplish the requirements of 009-032 of 2.1, including Table 3, Line(s) 20, located in 1.2.1 through 1.2.4, for coating removed in 5.3.

5.7.1 Power tool clean to bare metal for each area. Accomplish each requirement of Surface Preparation Specification SSPC-SP 11 of 2.8.

5.8 Accomplish the requirements of 009-032 of 2.1 for new and disturbed surface.

5.9 Notes:

5.9.1 Planner: Code 231A M. Kanehira, 243-9883
Planner: Code 231A R. Nanba, 243-6141

5.9.2 Work Integration Manager (WIM): Code 333 T. Yaginuma, 080-6522-9560.

5.9.3 This TGI meets the intent of DDG CSWT File No. 251-043.

5.9.4 Work per this instruction involves cutting an access (or removal of a door) through a fire zone bulkhead, through the hull of the ship above the turn of the bilge, through ventilation components (ducting, plenums, etc.), or through any air-tight, fume-tight or watertight bulkhead normally used to control air flow in the event of a shipboard fire. A non-flammable closure is required in accordance with COMNAVDEASYS COM MSG R 261400Z JUL 12. The closure, if not continuously in use, shall be staged near the access, readily visible and accessible, so that any emergency responder can install the closure to affect proper air flow onboard the ship.

5.9.5 Removal and installation of interferences, and gas free operation, preservation of boundary spaces relating to front loading structure repair are not included in this TGI. They will be covered by a DL in the future based on Contractor submitted condition report.

5.9.6 Materials for structure members relating to front loading repair will be provided in the future as a DL based on Contractor submitted condition report.

5.9.7 WPC will copy the following pages in reference.

REF. 2.5: Sheets 6-2, 3-6, 6B-6, 5-8, and 5D-8.

REF. 2.7: Sheets 5-9, 3-10, and 7-10.

5.9.8 JCNs EM01-ZCJ5, EM01-ZCJ6, EM01-ZCJ7, and EM01-ZCJ8 are related to this TGI.

5.10 Government Furnished Material (GFM): None

5.11 SRF-JRMC will provide the following service work: (TGI No. 38KB312388S02)

5.11.1 [C740] Provide crane and crane rigging service.

5.11.2 [X99E] Provide utility service, electrical.

5.11.3 [X99M] Provide utility service, including low pressure air supply, water supply, and steam supply.

5.11.4 [C106.1] Provide safety inspection for SRF-JRMC employees.

5.11.5 [C106.2] Provide gas free inspection for SRF-JRMC employees.

5.11.6 [C133] Provide QA inspection for (G) symbol(s).

5.11.7 [C250] Provide on-site engineering support for front loading.

FWD INTAKE	AY	A	0
DDG-76	CHG/VER:	0/0	

5.11.8 [C285] Provide inspection for (G) symbol(s).

6	TGI NO.:	FWD INTAKE	AY	A	0
	SHIP:	DDG-76	CHG/VER:		0/0

APPENDIX "J"

TGI JOB SUMMARY INFORMATION SHEET

SYSTEM:	PLANT NO.:	COMPONENT NAME: FWD INTAKE	TYPE: AY	PHASE: A	REPEAT: 0	CU PH VERSION: 0	CASREP CAT:
CU PHASE TITLE: F/O – INSPECT, REPAIR, AND PRESERVE IN NO. ONE GTM AND SSGTG NO. ONE INTAKE							
PRIMARY COMPONENT:							
COMPONENT UNIT LIST (FOR ASSEMBLIES): #1B GTM INTAKE IT CMPT 02-174-0-Q MS CMPT 03-174-0-Q MS CMPT 02-161-2-Q MS #1A GTM INTAKE IT CMPT 02-174-1-Q MS							
ATTACHMENT LIST:							
ZONE MANAGER: YAGINUMA ,TAKEHIDE		ZONE:	COMPARTMENT:	LEVEL:	FRAME:	SIDE:	
JO: 38KB312388	KO: A01	TOTAL MAN HRS:	TOTAL MAT'L COST	TOTAL DURATION:	PLANNING ACCEPTANCE:		
CU PHASE DESCRIPTION: The purpose of this TGI is to inspect, repair, and preserve in engine room No. One gas turbine module (GTM) and ship service gas turbine generator (SSGTG) No. One dirty side intake Location of Work: Intake Engine Room No. One, Port (Dirty Side) (02-174-0-Q) Intake Engine Room No. One, Stbd (Dirty Side) (02-174-1-Q) SSGTG Intake (Dirty Side) (03-174-0-Q) SSGTG Intake/ Exhaust Trunk (Dirty Side) (02-161-2-Q) SSGTG Intake Trunk (2-161-2-Q) Tomahawk Equipment Room (2-153-2-C) SSGTG Intake/ Exhaust Trunk (Clean Side) (02-161-2-Q) Intake Engine Room No. One, Port (Clean Side) (02-174-0-Q) Intake Engine Room No. One, Stbd (Clean Side) (02-174-1-Q) Auxiliary Machinery Room No. One (4-126-0-E) Engine Room No. One (4-174-0-E)							

J-1 TGI NO.:

SHIP:

FWD INTAKE	AY	A	0
DDG-76	CHG/VER:		0/0

J-2	TGI NO.:	FWD INTAKE	AY	A	0
	SHIP:	DDG-76	CHG/VER:		0/0

APPENDIX “M”
TGI MATERIAL LIST

SYSTEM:	PLANT NO.:	COMPONENT NAME:	TYPE:	PHASE:	REPEAT:	CU PH VERSION:	CASREP CAT:
		FWD INTAKE	AY	A	0	0	

CU PHASE TITLE:
F/O – INSPECT, REPAIR, AND PRESERVE IN NO. ONE GTM AND SSGTG NO. ONE INTAKE

MATERIAL REFERENCES			
REF NO.	DOCUMENT/DRAWING NUMBER	DOCUMENT/DRAWING TITLE	REV

JOB ORDER:	KEY OP:
38KB312388	A01

MATERIAL LIST

ITEM NO.	PC NO.	REF NO.	M / C	DESCRIPTION	STOCK NUMBER	GENERIC MATL TYPE	LVL	QA	HAZ MAT	DOC NO.	QTY REQD/ UNIT	QTY USED/ UNIT
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NOTES:

M-1

TGI NO.:	FWD INTAKE	AY	A	0
SHIP:	DDG-76	CHG/VER:	0/0	

☐ NUCLEAR

☐ NON-NUCLEAR

APPENDIX "D"
DEFICIENCY FORM

1. SHIP/HULL NO.: DDG-76	2a. DL or DR No. (Circle One)	2b. REVISION:	3a. COMPONENT NAME: FWD INTAKE	3b. TYPE: AY	3c. PHASE: A	3d. REPEAT: 0
4a. ICN: 38KB312388	4b. KEY OP: A01	5. LOC SPACE/LVL:	FRAME:	P/S/C:	6. CAUSED BY:	
7. DESCRIPTION:						
8. ORIGINATOR:	NAME:	BADGE:	SHOP/CODE:	DATE:	PHONE NO.:	
9. CORRECTIVE ACTION:						
10. EVALUATOR:	NAME:	BADGE:	SHOP/CODE:	DATE:	PHONE NO.:	
FOR NON-NUCLEAR USE ONLY						
11. RFD/RFW REQ'D ☐		LAR REQ'D ☐		SRD/PLAN REV REQ'D ☐		
ILS MOD REQ'D ☐		DWG NO.: _____ S/A NO.: _____				
12. ISOLATION SATISFACTORY:	SIGN:	BADGE:	SHOP/CODE:	DATE:	PHONE NO.:	
13. SUBSAFE OQE RECORD? YES ☐ NO ☐	SIGN:	BADGE:	SHOP/CODE:	DATE:	PHONE NO.	
14. CORRECTIVE ACTION COMPLETE:	SIGN:	BADGE:	SHOP/CODE:	DATE:	PHONE NO.	
15. INSPECTION REQ'D? :YES ☐ NO ☐	INSPECTION COMPLETE/SIGN:	BADGE:	SHOP/CODE:	DATE:	PHONE NO.	
16. RECORDS REVIEW COMPLETE (DR ONLY)	SIGN:	BADGE:	SHOP/CODE:	DATE:	PHONE NO.	
DL's ISSUED (DR ONLY)						

